Amdad Ahmed Awsaf

786-###-#### | aawsa001@fiu.edu | Website | Google Scholar | LinkedIn

(Authorized to work in the United States; no sponsorship required now or in the future)

RESEARCH PROFILE

Education researcher with 7+ years of experience in teaching, quantitative and qualitative research, tool development, and program evaluation. Methods include linear and logistic regression, structural equation modeling, and mixed-methods integration. Research focuses on STEM participation, college major intentions, and student persistence; contributes to research reports, manuscripts, and grant proposals.

EDUCATION

PhD, STEM Education (ABD) ***Spring/Summer 2027***
Department of Teaching and Learning, Florida International University, Miami, Florida
GPA 4.00/4.00; Dissertation advisor: Dr. Remy Dou

Education Specialist in Teaching and Learning, STEM Education ***2025***
Florida International University, Miami, Florida
GPA 4.00/4.00

Master of Education in Science, Mathematics, and Technology Education ***2022***
University of Dhaka, Dhaka, Bangladesh
GPA 4.00/4.00

Thesis: Nature and impact of informal professional learning communities' technological support during COVID-19 in enhancing secondary science teachers' efficacy

Bachelor of Education in Science, Mathematics, and Technology Education ***2019***
University of Dhaka, Dhaka, Bangladesh
GPA 3.80/4.00

RESEARCH METHODS AND ANALYTICS

Graduate methods preparation: 27 credits of research-methods coursework

- **Quantitative**: Introductory Data Analysis in Educational Research; Advanced Analysis in Educational Research; Structural Equation Modeling in Educational Research; Survey Research Methods
- **Qualitative**: Qualitative Foundations in Educational Research; Qualitative Research Data Analysis; Advanced Qualitative Research
- **Mixed and General Methods**: Seminar in Mixed Methodology; Engineering and Computing Education Research Methods in Context
- **Analytic expertise:** Linear and logistic regression; structural equation modeling; measurement invariance; survey and interview design; qualitative coding and phenomenological analysis; mixed-methods integration; systematic review
- **Software and platforms:** R (RStudio), SPSS, Microsoft Excel (proficient), Qualtrics, NVivo, MAXQDA, ATLAS.ti, Canva, Moodle

RESEARCH EXPERIENCE

Graduate Research Assistant, Florida International University (FIU) ***August 2025 – Present***
Advisor: Dr. Zahra Hazari, PhD

Research Projects: *STEP UP: Supporting Teachers to Encourage the Pursuit of Undergraduate Physics*

- Analyze quantitative and qualitative data on interventions supporting physics engagement, teacher professional learning, and students' disciplinary major intentions
- Contribute to research reports and manuscripts communicating findings on teacher learning, STEM participation, and student pathways
- Co-author grant proposals supporting ongoing and future research initiatives in physics education and STEM participation

Graduate Assistant, Florida International University (FIU) ***May 2026 – July 2026***
The DREAM Center
Advisor: Dr. Elizabeth Cramer

- Supported the federally funded OSEP technical assistance center focused on strengthening special education personnel preparation under IDEA
- Assisted with institutional reports and materials supporting program improvement, educator preparation, and dissemination
- Contributed to documentation and synthesis of center activities for training and technical assistance initiatives

Graduate Research Assistant, Florida International University (FIU) ***August 2023 – August 2025***
Advisor: Dr. Remy Dou, PhD

Research Projects:

1. *CAREER: Talking Science: Early STEM Identity Formation Through Everyday Science Talk" (NSF Award AISL #1846167)*
2. *OYE Tumble: Originating Youth Engagement with Culturally-Situated Science Podcasts for Kids (AISL #2415575)*
    - Examined youths' STEM identity development, emphasizing community cultural capital and everyday science discourse in informal learning contexts
    - Designed and refined surveys and semi-structured interview protocols using theory-driven research questions and stakeholder input
    - Conducted stakeholder and participant interviews and supported field-based data collection in informal and community STEM learning settings
    - Transcribed, managed, and coded confidential interview data in accordance with human-subjects research protocols and ethical standards
    - Analyzed and integrated qualitative and quantitative data using RStudio to examine patterns in STEM engagement, recognition, and identity development
    - Mentored and trained two undergraduate research assistants in data collection, transcription, analysis, and research procedures
    - Prepared research reports, conference presentations, and project deliverables communicating findings from NSF-funded research

Graduate Research Assistant, Center for Astrophysics | Harvard & Smithsonian
Advisor: Dr. Philip M. Sadler, Dr. Remy Dou, Dr. Sue Sunbury ***December 2022 – July 2024***
Research Project: *Racially and Ethnically Minoritized Youths' Out-of-School-Time Experiences and Their Effects on STEM Attitudes, Identity, and Career Interest* (NSF Award AISL #2215050)

- Researched the impact of community cultural capital on STEM engagement
- Designed surveys, conducted interviews, and analyzed quantitative and qualitative evidence on informal STEM learning experiences
- Transcribed interviews and managed confidential research data under established research protocols
- Assessed informal education programs and communicated findings to inform policy decisions

Early Career Researcher, University of Dhaka/The Open University UK ***March 2022 – July 2022***

Research Project: *3Mpower-Mobile Learning for Empowerment of Marginalised Mathematics Educators* research project

- Conducted field visits and teacher interviews for an international research collaboration on mathematics educators' professional learning
- Collected and analyzed qualitative data on professional learning experiences

Research Assistant (RA), University of Dhaka ***February 2022 – June 2022***

- Conducted faculty and student interviews for research on stakeholders' experiences of alternative assessment in higher education during the pandemic
- Conducted surveys and analyzed qualitative and quantitative data on stakeholders' experiences of higher education assessment

TEACHING AND MENTORING

Instructor of Record, Florida International University (FIU) ***Spring, 2026***
SMT4664 Educational Assessment and Project-Based Instruction in Mathematics and Science

- Led instruction and assessment for Florida Educator Accomplished Practices (FEAP)-aligned teacher certification course focused on STEM assessment and project-based instruction
- Co-designed curriculum and performance-based assessments with faculty mentor
- Evaluated 11 pre-service teachers
- Supervised 5 pre-service teachers in school-based practica and conducted site visits to 3 high schools for instructional evaluation

Graduate Teaching Assistant, Florida International University (FIU) ***Fall, 2022***
EDF 7404 Qualitative Research Data Analysis

- Supported a graduate qualitative data analysis course, including coding practice, analytic reasoning, and interpretation of findings
- Supported instructional activities on qualitative interpretation and communication of findings under faculty supervision

Lecturer of Higher Secondary Chemistry and Physics
Mission Academic Care, Dhaka, Bangladesh ***January 2017 - December 2020***

- Taught higher-secondary chemistry and physics to students preparing for advanced academic pathways.
- Developed lesson plans, instructional materials, classroom activities, and assessments aligned with subject standards and learner needs.

- Explained complex STEM concepts through accessible examples, guided practice, and technology-supported learning resources.

Intern Teacher, Udayan Higher Secondary High School, Dhaka ***January 2019 – June 2019***

- Instructed mathematics and science at the secondary level (middle and high school equivalent)
- Developed and implemented subject-based lesson and unit plans aligned with curriculum standards
- Assessed student learning through grading and evaluation of examinations and project-based work

Prepared teaching areas: qualitative and mixed-methods research; survey design; educational measurement and assessment; research data analysis; STEM education and teacher learning.

CURRICULUM AND INSTRUCTIONAL DESIGN

Curriculum Development Intern, BRAC Education Program, Dhaka ***October 2021-February 2022***

- Worked with the content team to research best practices and training in content development
- Reviewed books and other materials for new content and curriculum development
- Assisted the training team to conduct the assessment, evaluation, and follow-up with trainees

STEM Content Localizer, Agami Education Foundation, Dhaka ***March-December 2018***

- Localized science and mathematics content for the Khan Academy Bangla project, adapting digital learning resources for Bangla-speaking learners
- Built early experience with curriculum review, content alignment, learner accessibility, and culturally responsive resource development

MANUSCRIPTS UNDER REVIEW AND REVISION

Awsaf, A. A., Dou, R., Cian, H., & Secules, S. (2026). A Systematic Literature Review of Survey Research and Measurement of Engineering Identity. *European Journal of Engineering Education* (***under review***)

Awsaf, A. A., Hazari, Z., Sabouri, P., Mateu, C. (2026). Examining how teachers' disciplinary science identity and formal/informal professional development experiences relate to their students' disciplinary major intentions: A physics exemplar. *Journal of Science Teacher Education* (***under review***)

Awsaf, A.A., Siddique, M., N., A., Rahman, S., M., H. (2026). Examining the Impact of Informal PLCs' Technological Support on Science Teachers' Efficacy: An Exploratory Sequential Mixed Method Study. *The Electronic Journal for Research in Science & Mathematics Education* (***Revised and Resubmitted***)

Nti-Asante, E., **Awsaf, A. A.** (2025). *Constructing* Hybrid Mathematical Discourse: A Third-Space Approach to Ghanaian Dialects and English. *African Journal of Research in Mathematics, Science and Technology Education* (***Revised and Resubmitted***)

MANUSCRIPTS IN PREPARATION

Awsaf, A.A., Dou, R., Gerhard, S., & Sadler, P. (2026). Pre-College Recognition Experiences and STEM Identity Development Among College Youths: An Extended Structural Equation Model. *International Journal of STEM Education* (***in preparation***)

Awsaf, A.A., Chelsea, M., Hazari, Z., Sabouri. P., & Dreyfuss, B. B. (2026). Examining the Importance of Communal Classroom Culture for Students' Persistence in Physics and Engineering. *The Physics Teacher (**in preparation**)*

CONFERENCE PRESENTATIONS (PEER-REVIEWED)

Awsaf, A.A., Dou, R., Sunbury, S., Gerhard, S., & Sadler, P. (2026, April). *Carrying Contextual Learning Experiences Forward: A Structural Equation Model for Youths' STEM Identity Development* [Presentation]. 2026 AERA Annual Meeting, Los Angeles, California.

Awsaf, A.A., Dou, R., Sunbury, S., Gerhard, S., & Sadler, P. (2026, April). A Structural Equation Model for Youths' STEM Identity Development: Impact of Contextual STEM Learning Experiences [Presentation]. 2026 NARST Annual International Conference, Seattle, Washington.

Awsaf, A.A., & Irwin, C. (2026, April). Testing Cross-National Measurement Invariance of TIMSS 2023 EAB Framework and Implications in Environmental Education Research [Presentation]. 2026 NARST Annual International Conference, Seattle, Washington.

Awsaf, A.A., Dou, R., Gerhard, S., & Sadler, P. (*2025*, *April*). *Examining the impact of perceived experiences on STEM identity and career aspirations among underrepresented youth* [Presentation]. 2025 AERA Annual Meeting, Denver, CO.

Awsaf, A.A., Giasanti, N., Sunbury, S., & Dou, R. (*2025*, *March*). *Concurrent phenomenological analysis of STEM career aspirations in underrepresented youth: Role of experiences and identity* [Presentation]. 2025 NARST Annual International Conference, National Harbor, MD.

Awsaf, A.A., Dou, R., Gerhard, S., & Sadler, P. (*2025*, *March*). *Exploring associations between event-based (mis)recognition by STEM authorities with STEM identity and career aspirations* [Presentation]. 2025 NARST Annual International Conference, National Harbor, MD.

Awsaf, A.A., Cian, H., & Dou, R. (2024, March). *A Systematic Literature Review of Survey Research on Engineering Identity* [Presentation]. 2024 NARST Annual International Conference, Denver, CO.

Awsaf, A. A., Siddique, M. N. A., & Rahman, S. M. H. (October 2023). *Impact of Informal PLC's Technological Support During COVID-19 in Enhancing Secondary Science Teachers' Efficacy*. Southeastern Association for Science Teacher Education Annual Conference, Miami, Florida.

OTHER PRESENTATIONS AND INVITED TALK

1. Kim, J., **Awsaf, A.A.**, & Irwin, C. (2026, April). Testing Cross-National Measurement Invariance of TIMSS 2023 EAB Framework and Implications in Environmental Education Research [Presentation]. Graduate Student Appreciation Week (GSAW) Scholarly Forum 2026, FIU, Miami, FL.

2. **Awsaf, A.A.**, Dou, R., Sunbury, S., Gerhard, S., & Sadler, P. (2026, April). A Structural Equation Model for Youths' STEM Identity Development: Impact of Contextual STEM Learning Experiences [Presentation]. Graduate Student Appreciation Week (GSAW) Scholarly Forum 2026, FIU, Miami, FL.

3. **Awsaf, A.A.**, Dou, R., Gerhard, S., & Sadler, P. (2025, April). STEM Identity and Career Aspirations of College Youths: The Role of Event-Based (Mis)Recognition by STEM Authorities. *Poster presented at SEHD conference*, Miami, FL.

4. **Awsaf, A.A.** (2025, January). NISE Net Online Workshop: STEM Identity & Career Interest for Students Traditionally Underrepresented in STEM. National Informal STEM Education Network, Virtual, https://vimeo.com/1048979874?fl=pl&fe=cm.

HONORS AND AWARDS

2026 Jhumki Basu Fellowship Award
2026 Second Place, FIU Graduate Student Appreciation Week Scholarly Forum
2026 FIU Department of Teaching and Learning travel support
2026 CASE Travel Support
2025 NARST Graduate Student Committee Travel Scholarship
2025 FIU Graduate & Professional Student Committee (GPSC) Travel Support
2025 FIU Department of Teaching and Learning travel support
2025 CASE Travel Support
2024 FIU Graduate & Professional Student Committee (GPSC) Travel Support
2024 FIU Department of Teaching and Learning travel support
2024 CASE Travel Support
2022 Bangladesh Government Scholarship, Ministry of Education, Bangladesh

PROFESSIONAL MEMBERSHIPS

- National Association for Research in Science Teaching (NARST)
- American Educational Research Association (AERA)

ADDITIONAL PROFESSIONAL SERVICE

Examination Invigilator, British Council Dhaka ***December 2020 – May 2022***

- Conducted the A-level and O-level exams of Pearson, Edexcel, and Cambridge
- Invigilated in the IELTS exam

MISCELLANEOUS

Computer skills: R (RStudio), SPSS, Microsoft Excel, Qualtrics, NVivo, MAXQDA, ATLAS.ti, Moodle, Canva
Language skills: Bangla, English, Spanish (elementary proficiency)

ResearchGate: https://www.researchgate.net/profile/Amdad-Ahmed-Awsaf
ORCID: https://orcid.org/my-orcid?orcid=0009-0000-6943-4526